Analytic calculator for determination of γ-ray angular distribution coefficients and tensors in aligned and partially-aligned nuclei

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Abstract

A program has been developed to calculate a complete set of precise γ-ray angular distribution coefficients and statistical tensors in maximally- and partially-aligned nuclei. For practical nuclear structure and reaction purposes, there is no imposed constraint on any arguments that are likely to arise in the determination of these quantities. The program can also be used as a stand-alone vector-coupling calculator for the exact evaluation of Clebsch-Gordan and Racah coefficients, in addition to the related Wigner 3-j, 6-j, and 9-j symbols, that frequently arise in quantum mechanical applications involving angular momentum coupling and recoupling schemes.

Keywords: γ-ray angular distributions; angular momentum; Clebsch-Gordan coefficients; Wigner coefficients; 3-j symbol; Racah coefficients; 6-j symbol; 9-j symbol; nuclear structure

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PROGRAM SUMMARY

Program Title: PyGammaRAD: Python project for Gamma-Ray Angular Distributions
CPC Library link to program files: (to be added by Technical Editor)
Developer's repository link: `https://pypi.org/project/pygammarad/` and `https://github.com/AaronMHurst/PyGammaRAD`

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Code Ocean capsule: (to be added by Technical Editor)
Licensing provisions: MIT license
Programming language: `Python`
Nature of problem: Calculations based on angular momenta coupling schemes needed for the determination of γ-ray angular distributions coefficients and functions in maximally-aligned and partially-aligned nuclei, in addition to other quantum mechanical applications that require the exact evaluation of angular-momenta vector-coupling coefficients.
Solution method: Direct evaluation of sum formulae used to calculate angular momentum coupling coefficients and symbols, and the γ-ray angular distribution coefficients that are based on the determination of these quantities.

1. Introduction

In compound-nucleus reactions [1], particularly spectroscopy measurements involving heavy-ion induced fusion-evaporation reactions, the observed angular distribution of γ rays emitted from aligned states provides important nuclear structure information concerning multipolarity assignments, XL, where X denotes the electric E or magnetic M character and L represents the multipole order, of γ-ray transitions together with associated spin, J, and parity, π, assignments of the levels involved. The alignment condition requires that the magnetic substates m_i associated with a state of spin J have relative equal populations, i.e., $P_{m_i}(J) = P_{-m_i}(J)$. Experimentally, it is often found that the degree of alignment may only be partial and an appropriate attenuation factor will need to be deduced, e.g., in inelastic neutron-scattering measurements at relatively low neutron-induced energies $E_n < 10$ MeV [2]. However, in cases where the populated states are said to be well aligned and the emitted γ rays show a dependence on the transition multipolarities and spin sequence, resources including the *"Tables of Coefficients for Angular Distribution of Gamma Rays from Aligned Nuclei"* [3] and the *"Angular Distributions of Gamma Rays in Terms of Phase-Defined Reduced Matrix Elements"* [4] have long been valuable in the analysis and interpretation of experimental data. The published tables in Refs. [3, 4] include various statistical tensors and γ-ray angular-distribution coefficients, from which the overall γ-ray angular distribution function can be deduced, based on complete alignment of the magnetic substates. Collectively, this information permits for the determination of anisotropy coefficients for pure and mixed-multipole transitions involving states of definite J^{π} and where the γ-ray mixing ratio (δ_{γ}) for a given transition is known: $\delta_{\gamma} = 0$ for pure multipoles and $|\delta_{\gamma}| > 0$ for mixed transitions.

There are, however, practical limitations regarding the use of the data tables presented in Refs. [3, 4]. For example, the statistical population tensors in Ref. [3] are limited to ranges of $1 \leq J \leq 20$ for integral spins and $3/2 \leq J \leq 41/2$ for half-integral spins. Additionally, the angular-distribution coefficients listed in Ref. [3] are only calculated for transitions between states that differ by $\max(\Delta J) = 3$ and with corresponding multipole orders $1 \leq L \leq 3$. The highest spins involved have $J = 18\ \hbar$ and $J = 37/2\ \hbar$ for transitions between states of integral and half-integral spin sequence, respectively. There are similar limitations for the calculated tensors and angular distribution coefficients reported in Ref. [4] also. Although the tabulated results of Refs. [3, 4] will suffice for certain applications, they nevertheless represent a limitation where wider spin ranges and higher multipole orders are desired. In recent years, fusion-evaporation reactions at stable ion-beam accelerator facilities routinely produce many residual nuclei for a given beam-target combination populating rotational cascades to observational limits well beyond the spin values reported in Refs. [3, 4].

A further drawback of Refs. [3, 4] is the printed-form availability of the data, making their use inconvenient; extracting information from these articles may prove error prone. The purpose of this project is to provide an open-source toolkit capable of performing accurate floating-point calculations of quantities needed to describe γ-ray angular distributions in aligned nuclei over a much broader range of J and L. Not only does this serve as a check of the original works by Yamazaki [3] and Rose and Brink [4], but also removes the aforementioned limitations. Furthermore, the angular momentum coupling coefficients inherent to the calculation of the statistical tensors and angular distribution functions described in these works also have wider use in various branches of physics and chemistry involving vector-coupling problems based on the quantum theory of angular momentum [5, 6, 7]. Accordingly, it also intended for this project to provide a general-purpose stand-alone vector-coupling calculator allowing for accurate analytical determinations of Clebsch-Gordan and Racah coefficients in addition to Wigner 3-j, 6-j, and 9-j symbols. Although there are existing capabilities in this regard, this project offers a convenient self-contained package enabling straightforward execution of a complete set of angular-distribution coefficients and tensors in addition to vector-coupling calculations needed for most practical purposes. Existing capabilities only provide partial solutions for such needs. Furthermore, the program is designed to be small and written in a modular fashion based on a modern programming language that is easy to install and portable allowing for straighforward implementation in larger codebases.

A brief overview of the program structure is described in Sect. 2, along with details of where to obtain the project. The adopted formalism used for algorithm development in the calculation of quantities related to γ-ray angular distribution coefficients is described in Sect. 3. Sect. 4 provides a discussion of the the procedures used to benchmark the code including validation of evaluated quantities against known tabulated values available in the published literature. In Sect. 5 a summary and outlook is presented. Finally, in Appendix A we present an overview of the vector-coupling coefficients commonly used in quantum physics applications based on the theory of angular momentum.

2. Structure of the program

A Python implementation of a suite of calculators has been written to return precise analytic results for the γ-ray angular distribution coefficients defined in Sect. 3, together with the associated angular momentum coupling coefficients which are required in the determination of these quantities. The closely related angular momentum coupling symbols are also defined and coded into the suite, thus the complete software package offers broader application beyond the scope of the angular distribution focus of this manuscript. The program is structured in a modular fashion and a single source file `am_formulae.py` contains separate classes based on multiple inheritance for the angular momentum coupling coefficients: `am_formulae.ClebschGordan`, `am_formulae.Wigner3j`, `am_formulae.Racah`, and `am_formulae.Wigner9j` for the Clebsch-Gordan coefficient, Wigner-$3j$ symbol, Racah coefficient, and Wigner-$9j$ symbol calculators, respectively. The Wigner-$6j$ symbol is strongly related to the Racah coefficient through a simple phase factor (see Sect. Appendix A.5) and therefore a member function for its evaluation is also contained within the same `am_formulae.Racah` class. The calculation of the Wigner-$9j$ symbol is also simplified using `am_formulae.Racah` class-inherited methods, as described in Sect. Appendix A.6. The callable methods used to return the exact solution of these vector coupling coefficients are then contained within `AngularMomentumCalculations` class of the `am_methods.py` module.

The source file `angular_distributions.py` contains the appropriate member functions of the `AngularDistributions` class needed to calculate the γ-ray angular distribution coefficients and tensors listed in the tabulated data of Yamazaki [3], as well as those in the tables of the appendix in the review article by Rose and Brink [4]. To aid in the determination of the overall angular distribution calculation, the `angular_distributions.Legendre` class contains the first 11 polynomials of the Legendre series coded in terms of $\cos\theta$. Finally, the `tables.py` module contains classes and methods for handling and manipulating the tabulated data of Refs. [3, 4]. Effectively, this module serves as an API for interacting with the published tables in these works and also allows users to fully regenerate the tables in convenient machine-readable CSV and JSON formats.

2.1. Running the program

The `PyGammaRAD` software package is open source, freely available on GitHub [8] and the Python Package Index (PyPI) repository [9], and comes packaged with a permissive license. Building and installation are straightforward and are detailed in the accompanying `README` documentation on the GitHub landing page. The calculation of all quantities and utilization of associated functions is enabled by creating an instance of the `AngularMomentum` class after importing the `PyGammaRAD` library, for example:

```
In [1]: import PyGammaRAD as pg
In [2]: am = pg.AngularMomentum()

# Calculate Clebsch-Gordan coefficient <j1=5/2 m1=3/2
   j2=5/2 m2=-1/2 | j=1 m=1>
In [3]: am.cg(2.5, 1.5, 2.5, -0.5, 1, 1)
Out[3]: -0.4780914437337574 # -2*sqrt(2/35)

# Calculate coefficient F(k=2,Jf=0,L1=2,L2=2,Ji=2)
In [4]: am.calc_F(2, 0, 2, 2, 2)
Out[4]: -0.5976143046671966

# Compare to result in "Table 2(a)" [Yamazaki 1967]
In [5]: am.get_row_table2(2,0,2,2,2,coeff='F')
```

```
Out[5]: -0.59761

# Print "Table 2(a)" from Ref. [Yamazaki 1967] to file
   in JSON format
In [6]: am.table2file('T2A', 'JSON')
INFO:PyGammaRAD.log_handlers:YamazakiTable2a.json
```

To help illustrate various workflows and the overall utility of the software, including verification of methods against published results (see Sect. 4), `PyGammaRAD` comes shipped with different `Jupyter Notebooks` available from the project GitHub repository [8]. All `PyGammaRAD` classes and member functions have supporting docstrings that can be invoked at the console as described in the `README`. These docstrings provide a short explanation of the function, any arguments passed, the return value, as well as exceptions that are raised by the function. Each docstring is also presented with at least one known working example to showcase the corresponding method execution. All angular distribution and angular momentum coupling methods available to the `PyGammaRAD` library are also summarized in the associated `README` documentation, along with methods enabling retrieval and manipulation of the tabulated data in Refs. [3, 4].

2.2. Logging, testing, and dependencies

A logging framework has been implemented within `PyGammaRAD` whereby messages are categorized appropriately using different levels of severity. By default, logs are only directed to the console; however, an application log can also be redirected to file upon request. The implemented logging system provides a centralized and standardized way to handle the field reporting of application events such as errors and warnings. The logging system is configured to provide useful information and metadata regarding such events, including timestamps, module names, function names, and line numbers, thus enabling a richer context for debugging and analysis, making it easier for the user to monitor and troubleshoot issues.

A suite of Python modules comprising more than 200 unit tests has been developed as part of this project to help validate and benchmark its performance against published results and raising of exceptions when certain conditions arise. The test scripts are run to ensure compliance with the local Python environment and have been successfully tested across multiple

Python versions 3.6 to 3.13 inclusive on both Linux and Mac OS platforms. The `PyGammaRAD` software package has the following third-party Python-package dependencies: `numpy`, `pandas`, and `pytest`. There is also an additional `scipy` package dependency to run certain actions in the accompanying notebooks.

3. Formalism used in the calculations

The calculators used in the evaluation of the overall angular distribution coefficients and functions are based on the well-known sum formulae for the coupling of angular momenta vectors which have been extensively described in various dedicated texts on the subject matter, e.g., Refs. [6, 7, 5, 10], as well as in the published literature, e.g., Refs. [11, 12, 13, 14, 15, 16, 17, 18, 19, 20]. The vector-coupling routines are described in Appendix A, while the quantities explicitly of relevance to angular distributions − together with the adopted formalism − are described here and also summarized in the `README` documentation [8, 9] distributed with the `PyGammaRAD` library.

3.1. γ-ray angular-distribution coefficient calculator

Methods have been developed based on the formalism outlined by Yamazaki [3] and Rose and Brink [4] to calculate angular distribution functions in maximally-aligned nuclei. Additionally, we have further developed methods for partial alignment in nuclei based on the formalism outlined by Der Mateosian and Sunyar [21], which in itself is an extension of the earlier work by Yamazaki [3]. Here, we present an overview of the corresponding expressions used to calculate the various functions and coefficients that are needed to reproduce all the tabulated data in Refs. [3, 4, 21]. Furthermore, these methods can be applied for values of J, L, m, and k well beyond the ranges presented in the original works of Yamazaki [3], Rose and Brink [4], and Der Mateosian and Sunyar [21]. The adopted formalism based on complete and partial alignment are discussed, in turn, below.

3.1.1. Maximum nuclear alignment

The overall angular distribution function [3] for a γ-ray transition from an initial state to a final state $J_i \to J_f$ may be adequately expressed as

$$W(\theta) = 1 + A_2 P_2(\cos\theta) + A_4 P_4(\cos\theta), \tag{1}$$

where A_k are the anisotropy coefficients and P_k are the Legendre polynomials of order k. To obtain A_k, we first need to calculate the statistical population tensor B_k and the F_k angular distribution coefficient [3]. In the case of complete nuclear alignment, we can write the statistical population tensor for integral and half-integral spin, respectively, as

$$\begin{aligned} B_k(J) &= (-1)^J \sqrt{2J+1} \langle J0J0|k0\rangle; \\ B_k(J) &= (-1)^{J-\frac{1}{2}} \sqrt{2J+1} \langle J\frac{1}{2}J-\frac{1}{2}|k0\rangle, \end{aligned} \tag{2}$$

where $\langle J0J0|k0\rangle$ and $\langle J\frac{1}{2}J-\frac{1}{2}|k0\rangle$ are Clebsch-Gordan coefficients for given J and k. The F_k angular distribution coefficient is given as

$$\begin{aligned} F_k(J_f L_1 L_2 J_i) &= (-1)^{J_f - J_i - 1} \sqrt{(2L_1+1)(2L_2+1)(2J_i+1)} \\ &\times \langle L_1 1 L_2 -1|k0\rangle W(J_i J_i L_1 L_2; k J_f), \end{aligned} \tag{3}$$

where $W(J_i J_i L_1 L_2; k J_f)$ is a Racah coefficient, and L_1 and L_2 are the angular momentum multipole orders for a mixed transition whereupon the lowest two orders are considered such that the interfering multipole $L_2 = L_1 + 1$. In the case of a pure transition $L_1 = L_2 = L$. For the ideal case of maximal alignment we can then compute A_k using

$$\begin{aligned} A_k^{\max}(J_i L_1 L_2 J_f) &= \frac{1}{1+\delta_\gamma^2} \left[B_k(J_i) F_k(J_f L_1 L_1 J_i) + 2\delta_\gamma B_k(J_i) F_k(J_f L_1 L_2 J_i) \right. \\ &+ \left. \delta_\gamma^2 B_k(J_i) F_k(J_f L_2 L_2 J_i) \right], \end{aligned} \tag{4}$$

where δ_γ is the γ-ray mixing ratio. In the case of a pure transition when $L_1 = L_2$ there is no mixing and $\delta_\gamma = 0$. Alternatively, the anisotropy coefficient can also be described in terms of the closely related R_k angular distribution coefficient [4]:

$$\begin{aligned} R_k(L_1 L_2 J_i J_f) &= (-1)^{1+J_i-J_f+L_2-L_1-k} \sqrt{(2L_1+1)(2L_2+1)(2J_i+1)} \\ &\times \langle L_1 1 L_2 -1|k0\rangle W(J_i J_i L_1 L_2; k J_f). \end{aligned} \tag{5}$$

Clearly, Eqs. (3) for F_k and (5) for R_k differ only in their phases and can be related through the following expression

$$R_k(L_1 L_2 J_i J_f) = (-1)^{L_1 - L_2 + k} F_k(J_f L_1 L_2 J_i), \tag{6}$$

and so the anisotropy coefficient given in Eq. (4) may then be recast by replacing the F_k coefficients with their corresponding R_k coefficients [4], thus allowing for the overall distribution function (Eq. (1)) to be expressed in terms of R_k instead.

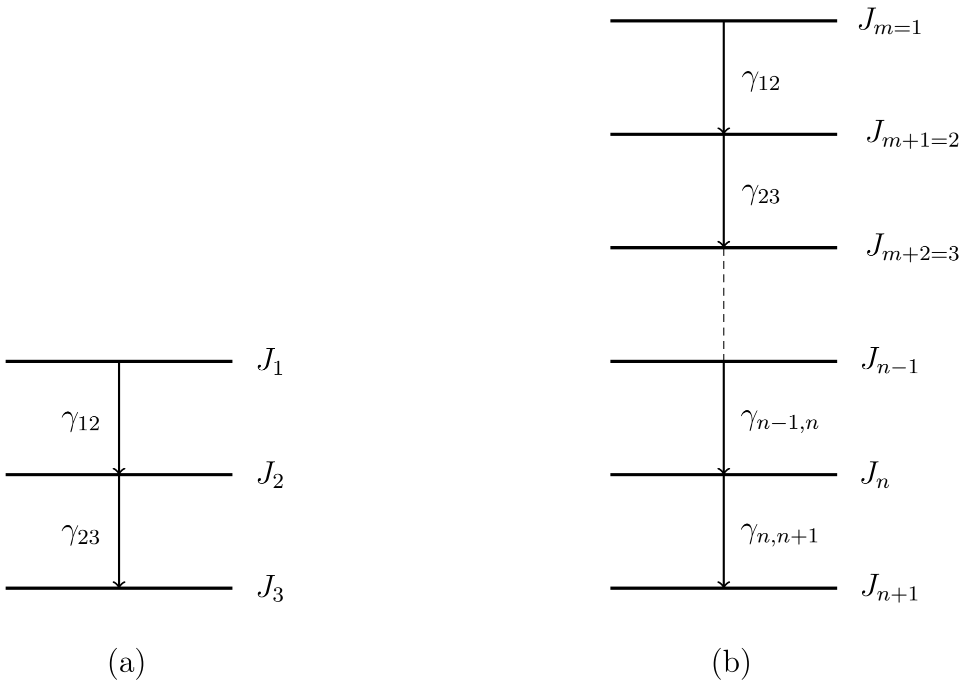

Figure 1: (a) Two-step γ-ray cascade. (b) Multi-step γ-ray cascade.

In a γ-ray cascade, for a state with spin J_f whose formation is dependent entirely upon the preceding $J_i \to J_f$ transition, it is useful to introduce the U_k angular distribution coefficient [3], often referred to as the deorientation (or depolarization) coefficient [22], which describes the effect of unobserved transitions preceding the observed radiation. This coefficient has the effect to alter the magnetic substate populations thereby reducing the degree of orientation ($U_k < 1$) and is dependent upon the multipolarities of the unobserved transitions, together with the angular momenta of the intermediary states, and may be written as

$$U_k(J_iL_1L_2J_f) = \frac{1}{1+\delta_\gamma^2}\left[u_k(J_iL_1J_f) + \delta_\gamma^2 u_k(J_iL_2J_f)\right], \tag{7}$$

where

$$u_k(J_iL_1J_f) = (-1)^{J_i+J_f-L_1}\sqrt{(2J_i+1)(2J_f+1)}W(J_iJ_iJ_fJ_f;kL_1). \tag{8}$$

Alternatively, Eq. (8) above may also be expressed as a ratio of Racah coefficients [4] given by

$$\begin{aligned} u_k(J_iL_1J_f) &= (-1)^k\frac{W(J_iJ_fJ_iJ_f;L_1k)}{W(J_iJ_fJ_iJ_f;L_10)} \\ &= (-1)^k\frac{W(J_iJ_iJ_fJ_f;kL_1)}{W(J_iJ_iJ_fJ_f;0L_1)}. \end{aligned} \tag{9}$$

If we consider the two-step γ-ray cascade involving transitions between states from $J_1 \to J_2 \to J_3$ as shown in Fig. 1a, where J_1 is the initially populated aligned state such that $P_{m_i}(J_1) = \frac{1}{2J_1+1}$ and therefore the normalization condition $\sum\limits_{m_i=-J_1}^{+J_1} P_{m_i}(J_1) = 1$ is satisfied, the distribution formula expressed in Eq. (1) will need to be modified for the second γ ray, labeled γ_{23} in Fig. 1a, which then has the following distribution [4]

$$W(\theta) = \sum_{k=0,2,4} U_k(J_i = J_1, J_f = J_2)A_k(J_i = J_2, J_f = J_3)P_k(\cos\theta). \tag{10}$$

Equation 10 can be generalized for the n^{th} γ ray in a multi-step cascade, as depicted in Fig. 1b, by including a product of U_k terms for each preceding non-observed γ ray, i.e., all γ rays other than $\gamma_{n,n+1}$ shown in Fig. 1b, such that

$$\begin{aligned} W(\theta) &= \sum_{k=0,2,4}\Bigg[A_k(J_i = J_n, J_f = J_{n+1})P_k(\cos\theta) \\ &\quad\times \prod_{J_m,J_{m+1}}^{J_{n-1},J_n} U_k(J_i = J_m, J_f = J_{m+1})\Bigg]. \end{aligned} \tag{11}$$

Another important quantity that arises in the treatment of γ-ray angular distributions is the statistical tensor. Rose and Brink [4] define this tensor in terms of both spin J and its quantized magnetic substate projection m, thus

$$\rho_k(J,m) = (2-\delta_{m,0})\sqrt{2J+1}(-1)^{J-m}\langle JmJ-m|k0\rangle, \tag{12}$$

where the Kronecker delta function $\delta_{m,0} = 1$ for $m = 0$; otherwise $\delta_{m,0} = 0$ for all other values of m. This expression removes the dependency of any knowledge of the associated population parameters [3], and accordingly, $\rho_k(J,m)$ can then be calculated for individual m-substate projections for a given J.

The final coefficient that we consider in this section is the S_k coefficient [4] which is written as

$$\begin{aligned} S_k(l_1l_2Js) &= (-1)^{s-J}\sqrt{(2l_1+1)(2l_2+1)(2J+1)} \\ &\times \langle l_10l_20|k0\rangle W(JJl_1l_2;ks), \end{aligned} \tag{13}$$

where l_1 represents the orbital angular momentum of the first partial wave and l_2 is that of the second interfering partial wave, and s denotes the reaction-channel spin angular momentum.

3.1.2. Partial nuclear alignment

In real experimental scenarios, the degree of alignment is often only partial and modifications to the overall distribution function given by Eq. (1) need to be accounted for. Here, an attenuation coefficient is defined such that

$$\alpha_k(J) = \frac{\rho_k(J)}{B_k(J)}, \tag{14}$$

where ρ_k represents the statistical tensor [3] for a state J defined by the population parameters P_m that specify the formation process for this state and is given by

$$\rho_k(J) = \sqrt{2J+1}\sum_{m=-J}^{J}(-1)^{J-m}\langle JmJ-m|k0\rangle P_m(J). \tag{15}$$

In cases where the alignment is only partial the states are oriented with relative unequal magnetic substate populations, i.e., $P_m(J) \neq P_{-m}(J)$, and $\rho_k(J)$ vanishes for odd-k. If $P_m(J)$ is unknown, then $\rho_k(J)$ given by Eq. (15) in the work of Yamazaki [3] is, in general, difficult to evaluate. Experimentally, however, it is often justifiable to assume a Gaussian distribution of magnetic substates [3, 21] characterized by the half-width parameter σ such that the m-substate population parameter may be calculated in accordance with the following expression

$$P_m(J) = \frac{\exp\left(-\frac{m^2}{2\sigma^2}\right)}{\sum\limits_{m=-J}^{J} \exp\left(-\frac{m^2}{2\sigma^2}\right)}. \tag{16}$$

Under this assumption, the dependence of the partial-alignment attenuation coefficient $\alpha_k(J)$ on the Gaussian-width distribution parameter σ/J can readily be demonstrated, and the modified angular distribution function may then be written as

$$W(\theta) = 1 + \alpha_2 A_2 P_2(\cos\theta) + \alpha_4 A_4 P_4(\cos\theta). \tag{17}$$

The $\alpha_k A_k$ term represents the attenuated anisotropy coefficient that can be compared with experimental anisotropy coefficients, frequently labeled a_k.

4. Verification of the programs

The methods presented in Sect. 3.1 were checked against the tabulated γ-ray angular distribution coefficients in the data tables of Yamazaki [3] and Rose and Brink [4]. In all cases, the published data are reproduced precisely, which further serves as an implicit verification of the calculation methods underpinning the evaluation of the Clebsch-Gordan and Racah coefficients that are needed in the overall determination of the various calculated angular-distribution coefficients. As a further check, the Clebsch-Gordan coefficients and Wigner 3-j, 6-j, and 9-j symbols defined in Appendix A were checked against all tabulated values published by Stevenson [19]. Additionally, we were able to confirm the calculation of these angular momentum couplings against the online Wigner coefficient calculator based on the root-rational-fraction program developed by Stone [15]. We can also assert that Eq. (8) [3] and Eq. (9) [4] are equivalent methods for determining the u_k angular distribution coefficient. The `Jupyter Notebooks` distributed with this project [8] contain actions confirming all the aforementioned published results, thus providing the user with convenient reference workflows and a means of inspection to perform independent checks of the data and calculation methods. Furthermore, as part of the test-driven development process, an extensive suite of unit tests has been written. These unit tests were, in part, developed to corroborate the well-established results of Refs. [3, 4, 19, 15]. It is intended for users to run through the test script to demonstrate the established conformity of all results within their local environment.

Some angular momentum couplings involving larger arguments were also confirmed against the results reported by Stevenson [19] and Johansson [20]. The built-in integer type of the Python programming language automatically handles arbitrarily large integers, so there is no inherent size limit beyond the available memory. However, for certain very large integers, for example, factorials of large numbers, the native `math.sqrt` method may not execute the conversion to a floating point representation, resulting in an `OverflowError` exception. To handle this exception, we instead perform an integer square root calculation based on Newton's iterative method [23] to refine an initial estimate until it converges with the correct integer square root. This `isqrt` method is a member function of the `am_formulae.Newton` class and has been tested against the native `math.sqrt` method using smaller integer arguments to ensure reproduciblity. Again, we provide a variety of unit tests and notebook actions to demonstrate the validity of the `am_formulae.Newton.isqrt` method. Finally, although we adopt the native `math.factorial` method and ensure arguments are interpreted as integers, we also performed checks against a user-defined recursive factorial method in addition to a Γ function where $\Gamma(n+1) = n!$. Consistency has also been demonstrated for these three methods, all of which are members of the `am_formulae.Factorial` class.

5. Summary

A set of calculators has been developed in Python to return precise γ-ray angular-distribution coefficients that are useful in the interpretation of nuclear structure data. These calculators have been benchmarked against the original works of Yamazaki [3] and Rose and Brink [4]. Although the range of arguments presented in the tables of these original works [3, 4] may be useful in many cases, for most practical purposes in nuclear structure our calculators are no longer constrained by the low limitations of these arguments and, e.g., calculations of coefficients involving nuclei populated to very high spin over a much wider range of multipolarities and magnetic substate projections may be readily evaluated. A further feature of this program allows users to regenerate the original tables of Refs. [3, 4] in convenient machine-readable

CSV and JSON formats. In particular, the defined vocabulary of the JSON schema provides implicit context regarding the nature and type of data held by the objects in their corresponding tables. These feature-rich formats, together with the associated contextual information described in this project, enable straightforward feature engineering allowing for mapping of raw data into feature vectors for further applications where labeled γ-ray angular distribution data may be useful. Extension of the tables to higher values of J, L, m, and k, as appropriate, can also be readily accomplished based on methods available within the `PyGammaRAD` library.

Because of the role of angular momenta in these calculations, the software package is also loaded with a stand-alone general purpose angular momentum coupling calculator enabling exact analytical evaluation of Clebsch-Gordan and Racah coefficients along with the Wigner 3-j, 6-j, and 9-j symbols. These angular momentum coupling calculators have been verified against the published works of Refs. [19, 15], and can thus find use in a broader range of quantum mechanical applications that involve the coupling of angular momenta. Finally, the small modular nature of `PyGammaRAD` allows it to be integrated into larger codebases where calculations of γ-ray angular distributions or angular momentum coupling coefficients are needed.

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Appendix A. Angular momentum coupling calculator

Although the development of a vector-coupling calculator is not a novel concept in itself, and indeed many such tools already exist that can satisfy this need, e.g., Refs. [11, 12, 13, 14, 15, 16, 17, 19, 18, 20, 24, 25], for convenience, the `PyGammaRAD` software package is also loaded with a complete angular momentum calculator that may serve a wide range of applications beyond the primary motivation of the work outlined in this paper. In the strict sense, it is only the Clebsch-Gordan and Racah coefficients that are needed to evaluate the angular distribution coefficients presented earlier in Sect 3.1. However, given their close relationship to the Wigner 3-j, 6-j, and 9-j symbols typically used to describe coupling and recoupling schemes involving angular momentum in a quantum mechanical context, to broaden the utility of the `PyGammaRAD` software package, we provide methods to readily evaluate all aforementioned angular momentum coefficients and symbols. The algorithms upon which these quantities are based are defined explicitly below.

Appendix A.1. Clebsch-Gordan coefficient

The Clebsch-Gordan coefficient describes the coupling of angular momentum vectors j_1 and j_2, each with magnetic substate projections m_1 and m_2, respectively, to form a system of total angular momentum j and corresponding projection m. This coupling scheme may be expressed as

$$\begin{aligned}\langle j_1 m_1 j_2 m_2 | j m\rangle &= \delta_{m,m_1+m_2}\Delta(j_1 j_2 j)\sqrt{(j+m)!(j-m)!(2j+1)} \\ &\times \sqrt{(j_1+m_1)!(j_1-m_1)!(j_2+m_2)!(j_2-m_2)!} \\ &\times \sum_{v=v_{\min}}^{v_{\max}}\left[\frac{1}{(j-j_2+m_1+v)!(j-j_1-m_2+v)!}\right. \\ &\times \left.\frac{(-1)^v}{v!(j_1+j_2-j-v)!(j_1-m_1-v)!(j_2+m_2-v)!}\right],\end{aligned} \tag{A.1}$$

where the triangular (or delta) factor is given by

$$\Delta(j_1 j_2 j) = \sqrt{\frac{(j_1+j_2-j)!(j_1-j_2+j)!(-j_1+j_2+j)!}{(j_1+j_2+j+1)!}}. \tag{A.2}$$

Only angular momentum vectors that satisfy the triangle inequalities condition result in the formation of a (j_1, j_2, j) triad arising from anti-parallel $(j_1 - j_2)$ and parallel $(j_1 + j_2)$ couplings:

$$|j_1 - j_2| \leq j \leq j_1 + j_2. \tag{A.3}$$

The summation over v in Eq. (A.1) runs over integers such that

$$\begin{aligned}v_{\min} &= \max(0, \max(-(j-j_2+m_1), -(j-j_1-m_2))); \\ v_{\max} &= \min(j_1+j_2-j, \min(j_1-m_1, j_2+m_2)).\end{aligned} \tag{A.4}$$

The phase convention for the Clebsch-Gordan coefficients evaluated using Eq. A.1 follows that of Condon and Shortley [26]. Additional conditions also apply regarding the magnetic quantum number projections: If j_i is integral then so must be all corresponding m_i projections, i.e., if $2j_i \bmod 2 = 0$ then $2m_i \bmod 2 = 0$. Likewise, for half-integral j_i and m_i projections, if $2j_i \bmod 2 = 1$ then $2m_i \bmod 2 = 1$. Also, for both integral and half-integral cases, each projection must satisfy the relation $|m_i| \leq j_i$. Finally, the sum of the coupling projections must combine to give the total angular momentum projection, i.e., $m_1 + m_2 = m$.

Appendix A.2. Wigner 3-j symbol

The Wigner 3-j symbol is based on the same sum formula used in Eq. (A.1) to calculate the Clebsch-Gordan coefficient and the two quantities are related through the following expression

$$\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m \end{pmatrix} = (-1)^{j+m+2j_1} \frac{1}{\sqrt{2j+1}} \langle j_1 - m_1 j_2 - m_2 | jm \rangle. \tag{A.5}$$

The same conditions regarding the magnetic substate projections for the Clebsch-Gordan coefficient also apply here in the calculation of the Wigner 3-j symbol, as does the triangle rule (Eq. (A.3)) for triad couplings. Note that for projections m_1, m_2, and m given in a Wigner 3-j symbol, the projections must couple as $(-m_1) + (-m_2) = m$ (i.e., $m_1 + m_2 + m = 0$).

Appendix A.3. Gaunt coefficient

Gaunt coefficients arise in vector-coupling calculations where a triple product of spherical harmonics integrated over a solid angle is needed, e.g., computation of molecular integrals based on Hartree-Fock-Roothaan methods [27]. Using the notation $\langle l_2 m_2 | l_1 m_1 | l_3 m_3 \rangle = Y^{l_3 m_3}_{l_1 m_1 l_2 m_2}$ [28], we may then express the Gaunt coefficient as

$$\begin{aligned} Y^{l_3 m_3}_{l_1 m_1 l_2 m_2} &= \int_{\phi=0}^{\phi=2\pi} \int_{\theta=0}^{\theta=\pi} Y_{l_1,m_1}(\theta,\phi) Y_{l_2,m_2}(\theta,\phi) Y_{l_3,m_3}(\theta,\phi) \sin\theta d\theta d\phi \\ &= \int_{\Omega=0}^{\Omega=4\pi} Y_{l_1,m_1}(\Omega) Y_{l_2,m_2}(\Omega) Y_{l_3,m_3}(\Omega) d\Omega, \end{aligned} \tag{A.6}$$

where $Y_{l,m}(\Omega)$ is a spherical harmonic defined by orbital angular momentum and magnetic quantum numbers l and m, respectively, and $d\Omega = \sin\theta d\theta d\phi$ is the infinitesimal element of solid angle subtended. This Gaunt coefficient, as given by the integral of Eq. (A.6), can be conveniently calculated as a generalization of two Wigner 3-j symbols using the following expression

$$\begin{aligned} G(l_1 l_2 l_3, m_1 m_2 m_3) &= \sqrt{\frac{(2l_1+1)(2l_2+1)(2l_3+1)}{4\pi}} \\ &\times \begin{pmatrix} l_1 & l_2 & l_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \end{aligned} \tag{A.7}$$

Gaunt coefficients are governed by selection rules and have non-zero values only if $m_1 + m_2 + m_3 = 0$, the triangle condition $|l_1 - l_2| \leq l_3 \leq l_1 + l_2$ is met, and $(l_1 + l_2 + l_3) \mod 2 = 0$.

Appendix A.4. Racah coefficient

The Racah coefficient $W(j_1 j_2 J j_3; J_{12} J_{23})$ involves the coupling of three angular momenta vectors. Here, there are two associated coupling schemes. In the first coupling scheme, j_1 couples with j_2 to form J_{12}, and then this resultant J_{12} vector couples with j_3 to form total angular momentum J, i.e., $|((j_1 j_2)J_{12}, j_3)J\rangle$. The other coupling scheme involves coupling j_2 with j_3 to give J_{23}, and j_1 next couples with J_{23} to give the same resultant total angular momentum J, i.e., $|((j_2 j_3)J_{23}, j_1)J\rangle$. Both coupling schemes result in complete orthonormal bases for the $(2j_1+1)(2j_2+1)(2j_3+1)$ dimensional space, and the scalar product of these two angular momentum bases yields the Racah recoupling coefficient. Explicitly, we may evaluate this coefficient as

$$\begin{aligned} W(j_1 j_2 J j_3; J_{12} J_{23}) &= \Delta(j_1 j_2 J_{12})\Delta(J j_3 J_{12})\Delta(j_1 J J_{23})\Delta(j_2 j_3 J_{23}) \\ &\times w(j_1 j_2 J j_3; J_{12} J_{23}), \end{aligned} \tag{A.8}$$

where each of the four triangular factors may be evaluated analogously to Eq. (A.2), subject to satisfaction of the triangle rule embodied by the inequality given in Eq. (A.3), and

$$\begin{aligned} w(j_1 j_2 J j_3; J_{12} J_{23}) &\equiv \sum_{z=z_{\min}}^{z_{\max}} \left[\frac{(-1)^{z+\beta_1}(z+1)!}{(z-\alpha_1)!(z-\alpha_2)!(z-\alpha_3)!(z-\alpha_4)!} \right. \\ &\times \left. \frac{1}{(\beta_1 - z)!(\beta_2 - z)!(\beta_3 - z)!} \right], \end{aligned} \tag{A.9}$$

where

$$\begin{aligned}
\alpha_1 &= j_1 + j_2 + J_{12};\\
\alpha_2 &= J + j_3 + J_{12};\\
\alpha_3 &= j_1 + J + J_{23};\\
\alpha_4 &= j_2 + j_3 + J_{23},
\end{aligned} \tag{A.10}$$

and

$$\begin{aligned}
\beta_1 &= j_1 + j_2 + J + j_3;\\
\beta_2 &= j_1 + j_3 + J_{12} + J_{23};\\
\beta_3 &= j_2 + J + J_{12} + J_{23}.
\end{aligned} \tag{A.11}$$

The summation over z in Eq. (A.9) is finite and extends over limits given by

$$\begin{aligned}
z_{\min} &= \max(\alpha_1, \alpha_2, \alpha_3, \alpha_4);\\
z_{\max} &= \min(\beta_1, \beta_2, \beta_3).
\end{aligned} \tag{A.12}$$

Appendix A.5. Wigner 6-j symbol

Both the Racah coefficients and Wigner 6-j symbols are a generalization of Clebsch-Gordan coefficients and Wigner 3-j symbols that arise in the coupling of three angular momenta. The Wigner 6-j symbol and the Racah coefficient are related through a phase factor given by

$$\begin{Bmatrix} j_1 & j_2 & J_{12} \\ j_3 & J & J_{23} \end{Bmatrix} = (-1)^{j_1+j_2+j_3+J} W(j_1 j_2 J j_3; J_{12} J_{23}). \tag{A.13}$$

However, because of their [higher number of] symmetry properties, see, e.g., Ref. [18] which illustrates the invariance under cyclic and anti-cyclic column permutation, Wigner 6-j symbols are often preferred as a more efficient means of storing the recoupling coefficients [18]. Clearly, the usual constraint of the triangle rule given by Eq. (A.3) also applies to angular momentum couplings involved in the evaluation of Wigner 6-j symbols.

Appendix A.6. Wigner 9-j symbol

The Wigner 9-j symbols are a generalization of Clebsch-Gordan coefficients, Wigner 3-j, and Wigner 6-j symbols arising from coupling four angular momenta. As with the Racah coefficient (or Wigner 6-j symbol), the coupling of angular momenta can be achieved in different ways. In this case, we have $j_1 \otimes j_2 = J_{12}$ and $j_3 \otimes j_4 = J_{34}$, before coupling these resultants $J_{12} \otimes J_{34}$ to give total angular momentum J, i.e., $|((j_1, j_2)J_{12}, (j_3, j_4)J_{34})J\rangle$. In the alternate coupling scheme, $j_1 \otimes j_3 = J_{13}$ and $j_2 \otimes j_4 = J_{24}$, before coupling $J_{13} \otimes J_{24} = J$, i.e., $|((j_1, j_3)J_{13}, (j_2, j_4)J_{24})J\rangle$. Again, both coupling schemes provide a complete orthonormal basis for the $(2j_1+1)(2j_2+1)(2j_3+1)(2j_4+1)$ dimensional space and the scalar product of the associated functions for these coupling schemes gives the Wigner 9-j symbol. For computational purposes and to help simplify the coding task, it is convenient to decompose the Wigner 9-j symbol into its triple product of constituent Wigner 6-j symbols [29, 30]:

$$\begin{aligned}\left\{\begin{array}{ccc} j_1 & j_2 & J_{12} \\ j_3 & j_4 & J_{34} \\ j_{13} & j_{24} & J \end{array}\right\} &= \sum_{h=h_{\min}}^{h_{\max}} (-1)^h (h+1) \qquad \text{(A.14)} \\ &\times \left\{\begin{array}{ccc} j_1 & j_2 & J_{12} \\ j_{34} & J & \frac{h}{2} \end{array}\right\} \left\{\begin{array}{ccc} j_3 & j_4 & J_{34} \\ j_2 & \frac{h}{2} & J_{24} \end{array}\right\} \left\{\begin{array}{ccc} J_{13} & J_{24} & J \\ \frac{h}{2} & j_1 & j_3 \end{array}\right\}.\end{aligned}$$

This representation then allows for calculation of the Wigner 9-j symbol using the same sum formulae of the `am_formulae.Racah` class (Eqs. (A.8)-(A.12)). The summation in Eq. (A.14) extends over all values of h that satisfy the triangle inequality theorem in the corresponding triangular factors, such that no factorial has a negative argument within the limits given by

$$\begin{aligned} h_{\min} &= h_{\max} \bmod 2; \\ h_{\max} &= 2\min(j_1 + J, j_2 + J_{34}, j_3 + J_{24}). \qquad \text{(A.15)}\end{aligned}$$

Due to its symmetry properties, any Wigner 9-j symbol is invariant under reflection through its diagonals. Therefore, the Wigner 9-j symbol in Eq. (A.14) has different equivalent representations, for example

$$\begin{aligned}\left\{\begin{array}{ccc} j_1 & j_3 & J_{13} \\ j_2 & j_4 & J_{24} \\ j_{12} & j_{34} & J \end{array}\right\} &= \sum_{h=h_{\min}}^{h_{\max}} (-1)^h (h+1) \qquad \text{(A.16)} \\ &\times \left\{\begin{array}{ccc} j_1 & j_3 & J_{13} \\ j_{24} & J & \frac{h}{2} \end{array}\right\} \left\{\begin{array}{ccc} j_2 & j_4 & J_{24} \\ j_3 & \frac{h}{2} & J_{34} \end{array}\right\} \left\{\begin{array}{ccc} J_{12} & J_{34} & J \\ \frac{h}{2} & j_1 & j_2 \end{array}\right\},\end{aligned}$$

whereupon the integer limits of h are also given by those in Eq. (A.15). Computational methods based on formulations of alternative algorithms, rather than a summation over a triple product of 6-j symbols, have also been proposed elsewhere, e.g., Refs. [16, 17].

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